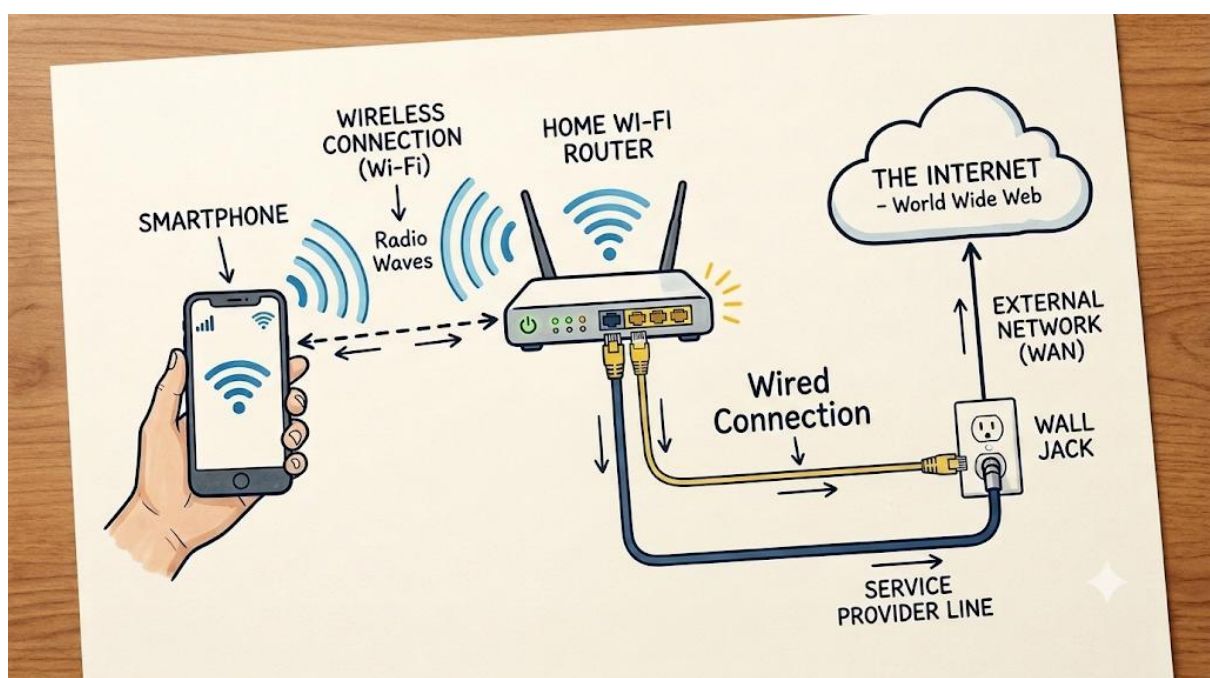


WIRELESS NETWORKING

1. POPULAR WI-FI STANDARDS & KEY DIFFERENCES :

- **802.11n (Wi-Fi 4):** OPERATES ON BOTH **2.4 GHz** AND **5 GHz** BANDS, OFFERING MAXIMUM THEORETICAL SPEEDS UP TO **600 MBPS**.
- **802.11ac (Wi-Fi 5):** OPERATES EXCLUSIVELY ON THE **5 GHz** BAND (FOR WI-FI TRANSMISSIONS), DELIVERING SIGNIFICANTLY HIGHER SPEEDS UP TO **3.5–6.9 GBPS**.
- **802.11ax (Wi-Fi 6):** OPERATES ACROSS **2.4 GHz, 5 GHz, AND 6 GHz (Wi-Fi 6E)** BANDS, BOOSTING MAXIMUM THEORETICAL SPEEDS UP TO **9.6 GBPS** WITH IMPROVED EFFICIENCY IN HIGH-DENSITY DEVICE ENVIRONMENTS.

2. NETWORK CONNECTIVITY TOPOLOGY :



3. Wi-Fi SCAN & NETWORK CHOICE :

WHEN YOU SCAN FOR Wi-Fi ON YOUR PHONE OR LAPTOP, YOU SEE A LIST OF NETWORKS:

- **HOME_NETWORK_5G:** STRONG SIGNAL, HIGH SECURITY (WPA3).
- **NEIGHBOR_GUEST:** WEAK SIGNAL, MEDIUM SECURITY (WPA2).
- **PUBLIC_CAFE_FREE:** MEDIUM SIGNAL, NO SECURITY (OPEN).

4. SSID HIDING EXPLAINED :

WHAT IS IT? AN SSID IS SIMPLY THE NAME OF YOUR Wi-Fi NETWORK. "SSID HIDING" MEANS SETTING YOUR ROUTER SO IT STOPS SHOWING ITS NAME IN THE LIST OF AVAILABLE NETWORKS. TO CONNECT, YOU HAVE TO TYPE IN THE EXACT Wi-Fi NAME AND PASSWORD MANUALLY.

- **ONE ADVANTAGE:**
 - **PRIVACY FROM NEIGHBORS:** CASUAL USERS OR NEIGHBORS WON'T SEE YOUR NETWORK LISTED WHEN SEARCHING FOR Wi-Fi.
- **ONE DISADVANTAGE:**
 - **NOT REAL PROTECTION:** FREE HACKER TOOLS CAN STILL EASILY SPOT HIDDEN NETWORKS. ALSO, YOUR PHONE WILL WASTE EXTRA BATTERY CONSTANTLY SEARCHING FOR THAT HIDDEN NAME WHENEVER YOU ARE AWAY FROM HOME.

